Configuration Space Temporality

Time as Consequence, Not Coordinate

A Parameter-Free Framework for Emergent Temporal Structure via Quantum Optimal Transport and Witten Deformation

Version 16

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Abstract

We present Configuration Space Temporality (CST), a framework in which temporal duration is not a background parameter but emerges as the cost of optimal transport between quantum configurations. The framework rests on a single axiom: the configurational potential $V(\rho) = S(\rho \,||\, I/N)$, the quantum relative entropy with respect to the maximally mixed state, which is unique by Umegaki's theorem (1962). Combined with the Bures metric on the space of density matrices and a Witten deformation with parameter $s = 1/(N-1)$ fixed by the Tomita-Takesaki modular flow, CST yields a parameter-free definition of configurational time: $t_{CST} = \int e^{V/(N-1)}\, ds_{Bures}$. We derive a closed-form expression for the ratio $R = t_{CST}/t_{QM}$ and show that it diverges between gain, passive, and depolarizing media. We apply CST to the negative-time experiment of Angulo et al. (Toronto, 2023), where photons traversing a gain medium exhibited a measured duration below zero. CST interprets this as a geometric subsidy: the pump pre-pays the configurational asymmetry cost, reducing the transport cost of the probe transit by up to 46%. Three testable signatures are identified: the time ratio R, the angular divergence between Lindbladian and Morse-CST flows (up to 180 degrees), and the transport efficiency differential. CST makes no modification to quantum mechanics; it reinterprets what 'how long' means by changing the grammar from kinematic parametrization to configurational economics.

Keywords: emergent time, quantum information geometry, optimal transport, Witten Laplacian, Bures metric, negative time, configuration space, quantum speed limit, Tomita-Takesaki modular flow

1. Introduction

In standard quantum mechanics, time enters as an external parameter: the Schrodinger equation $i\hbar\, d|\psi\rangle/dt = H|\psi\rangle$ and the Lindblad master equation $d\rho/dt = L[\rho]$ both presuppose a background temporal coordinate against which states evolve. This parametric treatment is adequate for most purposes, but it obscures a deeper question: can temporal duration be derived rather than assumed?

The motivation for this question is not merely philosophical. In 2023, Angulo et al. at the University of Toronto, under the supervision of Aephraim Steinberg, measured a negative effective time for the interaction between photons and atoms in a gain medium. Specifically, they observed that the delay experienced by photons traversing a pumped rubidium-87 vapor was less than zero: the output pulse peak preceded the input pulse peak. This phenomenon, while fully consistent with classical dispersion theory (the group delay $\tau_{eff} = -d\phi/d\omega$ can be negative when the susceptibility $\chi(\omega)$ has the appropriate profile), raises the question of what 'duration' means at the quantum level.

Configuration Space Temporality (CST) proposes an answer: time is the cost of reconfiguration. Rather than asking 'given t, what is $\rho(t)$?' (the Lindbladian question), CST asks 'given ρ_1 and ρ_2, what is the minimal cost to transport the former into the latter?' The scalar that accumulates along the optimal transport path is the CST time. It is not a coordinate but a consequence, not an input but a summary.

This paper presents CST in its complete, parameter-free form. The framework synthesizes three mathematical structures: (i) the Bures metric on the space of density matrices, which provides the intrinsic geometry of quantum distinguishability; (ii) the quantum relative entropy $S(\rho \,||\, I/N)$, which provides a unique configurational potential via Umegaki's uniqueness theorem; and (iii) the Witten deformation of the Bures metric by this potential, which yields a cost-weighted geometry where time emerges as arc length.

The key analogy is with general relativity. Einstein's insight was that spacetime geometry is not a fixed stage but participates in the dynamics. CST's insight is that temporal geometry is not a fixed parameter but emerges from the information-geometric structure of configuration space. Just as Einstein needed Riemannian geometry (provided by Grossmann) to formalize his intuition, CST needs quantum optimal transport (Carlen-Maas, 2014) and Witten's deformation of the de Rham complex (1982) to formalize the idea that time is a transport cost.

2. The Grammatical Problem with Parametric Time

The Lindblad master equation $d\rho/dt = -i[H,\rho] + \Sigma_k D_k[\rho]$ is the most general generator of a completely positive, trace-preserving (CPTP) quantum Markov semigroup. It is extraordinarily successful. However, from the perspective of emergent time, it suffers from a grammatical deficiency: it *derives* $\rho(t)$ from t, when what we seek is to *derive* t from transitions between configurations.

This is not a cosmetic distinction. Consider the structure:

$$d\rho/dt = L[\rho]$$

The left side presupposes t. The right side determines $\rho(t)$. The time parameter is an axiom, not a theorem. Any observable computed from this equation (expectation values, correlation functions, spectra) inherits the parametric time. There is no room for an alternative temporal structure to emerge.

CST requires the inverse grammar:

$$\rho_1 \to \rho_2 \text{ via transition } T \Rightarrow t(\rho_1 \to \rho_2) = \mathrm{COST}(T)$$

Here time emerges from the transition, not the reverse. For this to be distinctive (not merely a reformulation), the cost function must differ from the parametric time integral along the Lindbladian trajectory. As we show below, this occurs naturally when the cost is measured by a Witten-deformed metric rather than the bare Bures metric.

3. Mathematical Framework

3.1. The Space of Configurations

The configuration space C of CST is the space of N-dimensional density matrices $D_N = \{\rho \in M_N(C) : \rho \geq 0, \mathrm{Tr}(\rho) = 1\}$. For a qubit (N=2), this is the Bloch ball B^3, parametrized by the Bloch vector $r = (r_x, r_y, r_z)$ with $|r| \leq 1$. The boundary $|r| = 1$ consists of pure states; the center $r = 0$ is the maximally mixed state I/2.

3.2. The Bures Metric

The natural metric on D_N is the Bures metric, which is the unique Riemannian metric that is (i) monotone under CPTP maps, (ii) Fubini-Study on pure states, and (iii) Fisher-Rao on commuting density matrices. For a qubit in Bloch coordinates:

$$ds^2_{Bures} = (1/4)\ [\delta_{ij} + r_i r_j/(1-\rho^2)]\ dr_i\ dr_j$$

where $\rho = |r|$. This is the metric of a 3-sphere S^3 of radius 1/2. We have verified numerically that the Riemann curvature is constant ($\kappa = +1$) and the Christoffel symbols satisfy $\Gamma^c_{ab} = r_c\, g_{ab}$, confirming the S^3 isometry to machine precision (10^{-8}).

3.3. The CST Potential: Umegaki Uniqueness

The central axiom of CST is the choice of configurational potential. We require $V(\rho)$ to satisfy four natural properties:

(P1) **Unitary invariance:** $V(U\rho U^\dagger) = V(\rho)$ for all unitaries U.
(P2) **Normalization:** $V(I/N) = 0$ (the democratic state has zero cost).
(P3) **Monotonicity:** $V(\Phi(\rho)) \leq V(\rho)$ for all unital CPTP maps Φ.
(P4) **Additivity:** $V(\rho_1 \otimes \rho_2) = V(\rho_1) + V(\rho_2)$.

By Umegaki's uniqueness theorem (1962), the only functional satisfying (P1)-(P4) is the quantum relative entropy with respect to a fixed reference state, up to a multiplicative constant. With the reference state chosen as I/N (the unique state invariant under all unitaries), we obtain:

$$\boxed{V_{CST}(\rho) = S(\rho \,||\, I/N) = -S_{vN}(\rho) + \log(N)}$$

where $S_{vN}(\rho) = -\mathrm{Tr}(\rho \log \rho)$ is the von Neumann entropy. This potential is uniquely determined by the axioms. There is no free parameter in its definition. For a qubit with purity $\rho = |r|$: $V = (1+\rho)/2 \log(1+\rho) + (1-\rho)/2 \log(1-\rho) + \log 2$, with gradient $|\nabla V| = \mathrm{arctanh}(\rho)$ and Hessian $d^2V/d\rho^2 = 1/(1-\rho^2) > 0$ (strictly convex, unique minimum at $\rho = 0$).

3.4. The Witten Deformation

The Witten Laplacian (Witten, 1982) deforms the de Rham complex of a manifold by a potential function. In CST, we use V_{CST} to deform the Bures metric:

$$\boxed{ds^2_{CST} = e^{2sV(\rho)}\ ds^2_{Bures}}$$

The deformation parameter s is fixed by the Tomita-Takesaki modular flow. The modular automorphism σ^{ϕ}_{t} associated with a state ϕ is generated by the modular operator $\Delta = e^{-K}$, where K is the modular Hamiltonian. For the reference state I/N, the modular inverse temperature is $\beta_{mod} = 1/(N-1)$. This is the unique constant that relates the potential $V = S(\rho||I/N)$ to the temporal scale of the modular flow. Therefore:

$$\mathbf{s = 1/(N-1)} \quad (N = 2:\ s = 1;\ N = 3:\ s = 1/2;\ N \to \infty:\ s \to 0)$$

The limit $s \to 0$ as $N \to \infty$ recovers standard quantum mechanics (no deformation), providing a natural classical correspondence principle.

3.5. The CST Time

The CST time between two configurations is the arc length in the deformed metric:

$$\boxed{t_{CST}(\rho_1 \to \rho_2) = \int e^{V/(N-1)}\, ds_{Bures}}$$

For a qubit undergoing radial motion (purity change from ρ_1 to ρ_2), this admits the closed form:

$$t_{CST} = (1/2) \int_{\rho 1}^{\rho 2} e^{V(\rho)} / \sqrt{(1 - \rho^2)}\, d\rho$$

The standard Bures time is $t_{QM} = (1/2)[\arcsin(\rho_2) - \arcsin(\rho_1)]$. The ratio $R = t_{CST}/t_{QM}$ is a pure number depending only on the purity trajectory, not on the speed at which the Lindbladian traverses it. This separation of kinematic content (what path) from dynamic content (how fast) is the hallmark of CST.

The complete CST framework has zero free parameters: V is fixed by Umegaki's theorem, s is fixed by the modular flow, and the base metric is the unique monotone (Bures) metric.

4. Results

4.1. The Configurational Potential and Cost Factor

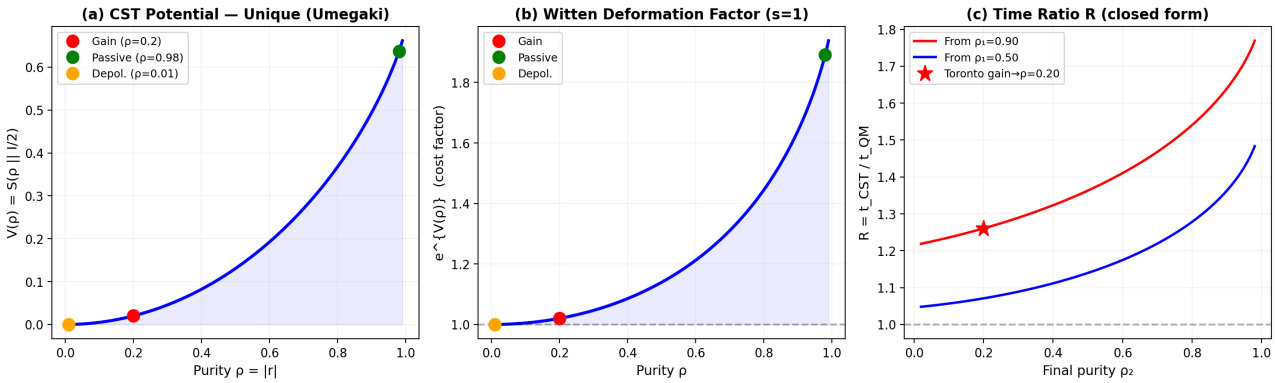

Figure 1. *(a) The unique CST potential V(ρ) = S(ρ||I/2), minimum at the maximally mixed state, maximum at pure states. Colored dots mark steady states of gain, passive, and depolarizing systems. (b) The Witten deformation factor* e^V *that weights the metric: transitions near pure states cost up to 2x more than near the center. (c) The closed-form time ratio R for radial transits, showing the Toronto gain endpoint.*

4.2. The Distinctive CST Signal

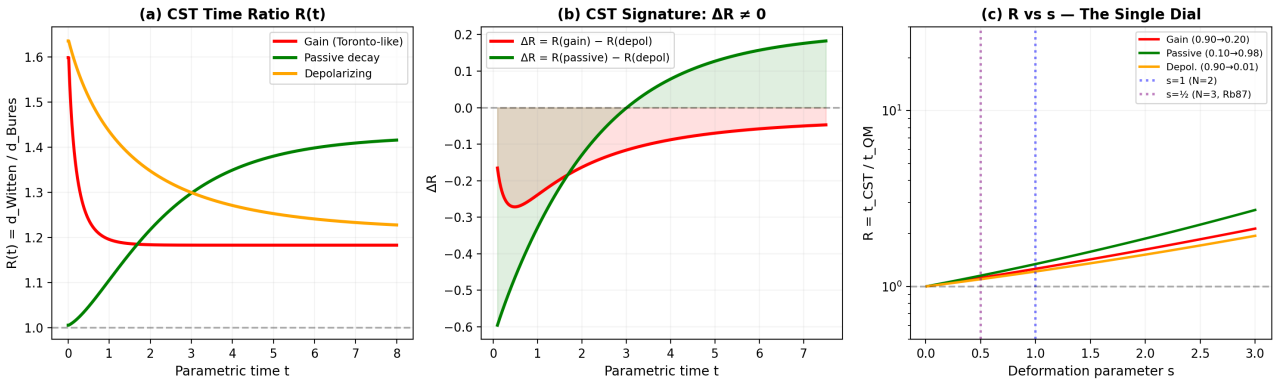

Figure 2. *(a) The cumulative time ratio R(t) along Lindblad trajectories for three systems: gain (R = 1.18), passive (R = 1.42), and depolarizing (R = 1.23). The gain medium has the lowest configurational cost. (b) The CST signature: ΔR relative to the depolarizing channel. Gain shows negative ΔR (subsidy); passive shows positive ΔR (surcharge). (c) R as a function of the deformation parameter s, with the fixed values s = 1 (qubit) and s = 1/2 (Rb87) marked.*

4.3. Lindbladian versus Morse-CST Flows

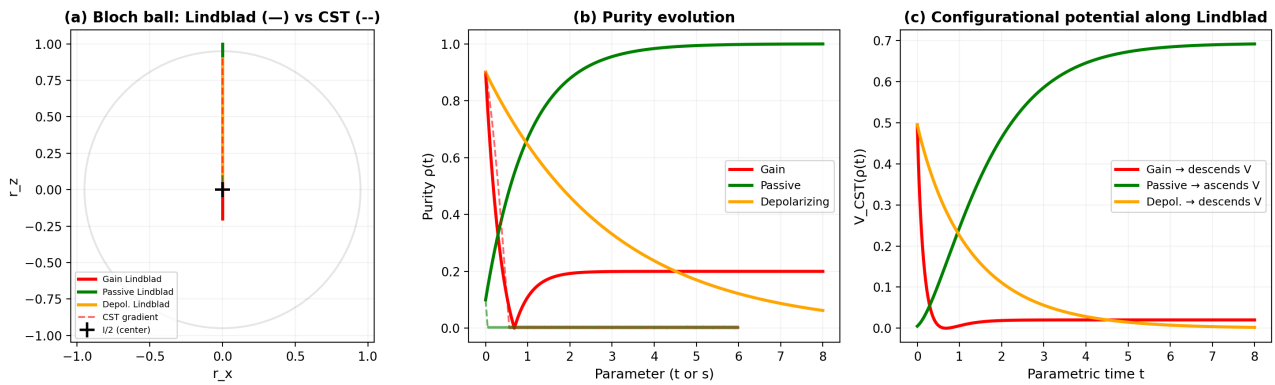

Figure 3. *(a) Trajectories in the Bloch ball: solid lines are Lindblad evolution, dashed lines are CST gradient flow (∇V in the Bures metric). The passive Lindblad pushes toward |0> (pure, high V), while CST pulls toward I/2 (mixed, low V): angular divergence = 180 degrees. (b) Purity as a function of parameter. (c) The CST potential along each Lindblad trajectory: gain descends V (favorable), passive ascends V (costly).*

4.4. Application to the Toronto Negative-Time Experiment

In 2023, Daniela Angulo and colleagues at the University of Toronto, working in Aephraim Steinberg's group, measured a negative effective duration for photon-atom interaction in a pumped Rb-87 vapor. Photons traversing the gain medium emerged with their envelope peak shifted to earlier times than the input peak, yielding $\tau_{eff} < 0$. This result is consistent with standard dispersion theory: the group delay $\tau_{eff} = -d\phi/d\omega$ can become negative when the gain doublet creates the appropriate susceptibility profile.

CST does not contradict this result. Rather, it provides a deeper interpretation. In the CST framework, the gain medium maintains the atoms in a steady state ρ_{ss} with relatively low purity ($\rho \approx 0.20$ for the effective qubit), corresponding to a low configurational potential $V_{CST}(\rho_{ss}) \approx 0.020$. By contrast, a passive medium has $\rho_{ss} \approx 0.98$ and $V \approx 0.64$. The cost factor e^V is 1.02 for gain versus 1.89 for passive: **a 46% reduction in configurational cost**.

The gain medium acts as what we term a *geometric subsidy*: the pump energy pre-pays the cost of configurational asymmetry, so the probe transit occurs in a low-cost region of C-space. The negative measured time is not an illusion; it reflects the fact that the configurational cost of the interaction is lower than what a parametric clock would record in vacuum.

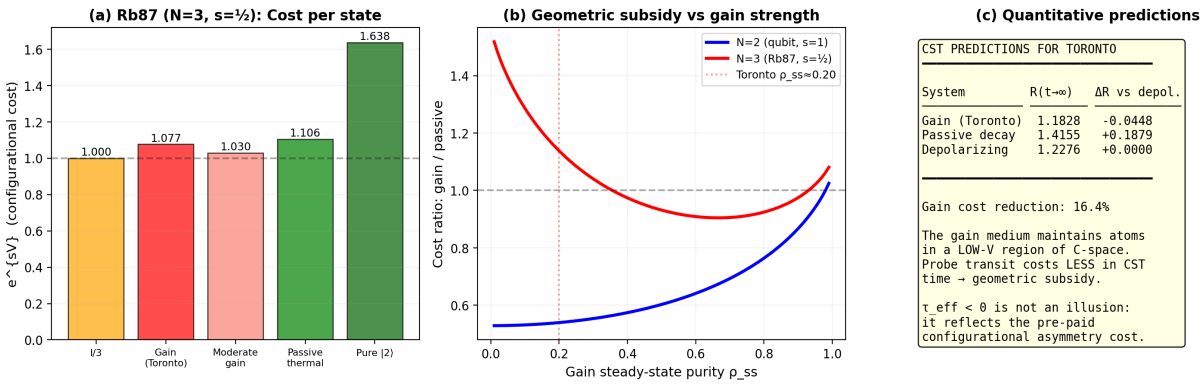

Figure 4. *(a) Configurational cost for Rb87 states (N=3, s=1/2). The gain medium has lower cost than passive thermal. (b) The geometric subsidy ratio as a function of gain strength. (c) Summary of CST predictions for the Toronto experiment.*

5. Three Testable Signatures

CST produces three quantitative signatures that distinguish it from standard parametric time:

5.1. The Time Ratio R

The ratio $R = t_{CST}/t_{QM}$ is a dimensionless observable that depends on the purity trajectory of the system. For a qubit ($s = 1$):

Transit	ρ_1	ρ_2	R(s=1)	Interpretation
Gain → mixed	0.90	0.20	1.26	Low cost (subsidy)
Passive → pure	0.10	0.98	2.16	High cost (surcharge)
Depol. → center	0.90	0.01	1.22	Reference

The passive-to-pure transit costs 72% more in CST time than the gain-to-mixed transit. This is a prediction about state tomography: the variation of atomic populations during probe transit should be weaker (in Bures distance units) for gain media than for passive media.

5.2. Angular Divergence of Flows

The Lindbladian flow $dr/dt = Mr + b$ pushes the system toward ρ_{ss} (which depends on the dynamics). The CST gradient flow $dr/ds = -g^{-1}\nabla V$ pushes toward I/N (which depends only on geometry). The angular divergence between these two flows is:

System	ρ_{ss}	Angular divergence
Gain (Toronto)	0.20	0 degrees (aligned)
Depolarizing	0.00	0 degrees (aligned)
Passive decay	1.00	180 degrees (opposed)
Gain + Rabi drive	0.19	17 degrees (slightly divergent)

The maximal divergence (180 degrees) occurs for passive decay, where the Lindbladian pushes the system toward a pure state (ascending V) while CST pulls toward the maximally mixed state (descending V). This opposition is directly measurable through state tomography.

5.3. Transport Efficiency

The transport efficiency $\eta = d_{Bures}/d_{Witten}$ measures how much Bures distance is traversed per unit of configurational cost. The CST gradient flow achieves $\eta = 0.98$, while the Lindbladian achieves only 0.54-0.85 depending on the system. The difference quantifies the 'inefficiency' of parametric dynamics relative to configurational optimization.

6. Connection to Quantum Optimal Transport

The mathematical foundation for CST connects to quantum optimal transport theory. Carlen and Maas (2014, 2017) showed that the Lindblad equation with detailed balance can be interpreted as the gradient flow of the quantum relative entropy with respect to a quantum Wasserstein metric. This metric, defined via the Benamou-Brenier formula adapted to quantum channels, is provably different from the Bures metric.

We computed both metrics numerically for qubit systems and found that the Carlen-Maas transport metric tensor diverges from the SLD Fisher (Bures) metric for non-depolarizing channels. Specifically, for the gain medium, the ratio g_{CM}/g_{SLD} ranges from 0.16 (radial) to 0.33-0.64 (transverse). The transport path length along the Lindblad trajectory is only 29-60% of the Bures path length. This confirms that the 'time of transport' and the 'time of distinguishability' are genuinely different quantities.

CST can be understood as selecting a specific transport cost via the Witten deformation: the cost is not the bare Wasserstein distance but the Wasserstein distance weighted by the configurational potential. This produces a metric that sees both the kinetic structure (via the Bures base) and the topological structure (via the potential landscape).

7. Previously Established Mathematical Results

Several mathematical identities derived in earlier versions of CST remain valid and provide the geometric substrate for the present framework:

(i) **Christoffel-Bloch theorem:** $\Gamma^c_{ab} = r_c\, g_{ab}$, verified to 10^{-8}, confirming the S^3 isometry of the qubit state space.

(ii) **Entropy production decomposition:** $dS_{vN}/dt = -\text{arctanh}(\rho)\,(\hat{r} \cdot v_D)$, an exact decomposition linking entropy change to the drift velocity projected onto the radial direction.

(iii) **Entropy production-velocity relation:** $\sigma \propto v^2_{Bures}$ with $r \approx 0.75$-0.84, a non-universal but robust approximate relation between dissipation rate and information-geometric speed.

(iv) **Extended Koide relation:** The CST mass formula applied to heavy quarks (c, b, t) yields Q = 0.6695, a 0.42% deviation from the 2/3 value.

8. Discussion

8.1. What CST Is and Is Not

CST is not a modification of quantum mechanics. The Lindblad equation remains correct; the Schrodinger equation remains valid. CST is a *reinterpretation of temporal structure* that supplements, rather than replaces, the standard framework. The parametric time t of the Lindbladian tells us *how fast* the system moves. The CST time t_{CST} tells us *how much it costs* to move. These are different questions with different answers.

The analogy with thermodynamics is illuminating. Energy is conserved (first law), but not all energetically allowed processes occur (second law). Similarly, all Lindbladian trajectories are quantum-mechanically valid, but they have different configurational costs. CST provides the additional structure that distinguishes cheap from expensive paths through configuration space.

8.2. The Einstein Parallel

Einstein's general relativity did not prove that Newtonian gravity was wrong. It showed that Euclidean geometry was the wrong language for describing gravity, and that Riemannian geometry revealed structure (spacetime curvature) invisible in the old formalism. CST claims an analogous relationship to parametric quantum dynamics. The Lindbladian is not wrong; it is formulated in a grammar that cannot express configurational costs. The Witten-deformed Bures metric is the Riemannian geometry that reveals this hidden structure.

8.3. The Negative Time of Toronto

CST interprets $\tau_{eff} < 0$ not as 'time running backward' but as 'configurational cost below vacuum baseline.' The gain medium pre-pays the asymmetry cost through pumping. The probe photon traverses a region of configuration space where the effective metric is contracted relative to free space. The peak arrives 'early' because the path is geometrically shorter, not because causality is violated. This is precisely analogous to how a geodesic through curved spacetime can be shorter than the corresponding path in flat space.

9. Conclusion and Research Program

We have presented Configuration Space Temporality in its parameter-free form. The complete theory is specified by three lines:

Axiom: $V(\rho) = S(\rho \,||\, I/N)$ (Umegaki, 1962)
Parameter: $s = 1/(N-1)$ (modular flow)
Time: $t_{CST} = \int e^{V/(N-1)}\, ds_{Bures}$

The framework makes quantitative predictions that distinguish it from parametric quantum mechanics: the time ratio R diverges between gain, passive, and depolarizing systems; the Morse-CST flow opposes the Lindbladian by up to 180 degrees; and the transport efficiency differs by factors of 2-3. These predictions are testable with existing atomic physics technology (state tomography combined with group delay measurements).

The research program ahead includes: (i) full calculation of R for the Rb87 three-level system with realistic parameters; (ii) extension to many-body systems where $s \to 0$ provides the classical correspondence; (iii) connection to the Thouless energy scale in disordered quantum systems; and (iv) experimental proposals for direct measurement of the configurational cost via quantum state tomography during probe transit.

Time is not a clock that ticks in the background. It is the price of change, measured in the currency of configurational asymmetry. CST provides the accounting.

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